

Week 01_Module 01_Part 02

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Mean (Average)

Mean is what we usually call the *average*.

You add up all values and divide by the number of observations

Formula:

$$\text{Mean} = \frac{\sum x_i}{n}$$

Example:

For [10, 12, 14, 16, 18, 100]

Mean = (10 + 12 + 14 + 16 + 18 + 100) / 6 = 28.3

Median

Median is the middle value when data is sorted.

It splits the dataset into two equal halves.

Example:

Sorted [10, 12, 14, 16, 18, 100] → middle two are 14, 16

Median = (14 + 16)/2 = 15

Mode

Mode is the most frequent value in the dataset.

Example:

For [2, 2, 3, 3, 3, 4, 5], mode = 3

Comparison & When to Use Which

Situation	Best Measure	Reason
Symmetric data (e.g., height,	Mean	All values contribute

temperature)		equally
Skewed data with outliers (e.g., income, price)	Median	Resistant to outliers
Categorical data (e.g., color, gender)	Mode	Works with non-numeric data

Common Mistakes

- Using mean with skewed data → misleading.
- Using median for categorical → meaningless.
- Using mode for continuous numeric → rarely helpful.
- Forgetting to check distribution before imputing missing values.

Summary & Wrap-Up

- **Mean** – good for normal, balanced data.
- **Median** – good for skewed or outlier-rich data.
- **Mode** – best for categorical data.

In Machine Learning, choosing the right measure helps you **represent data correctly**, **fill missing values wisely**, and **build more accurate models**.